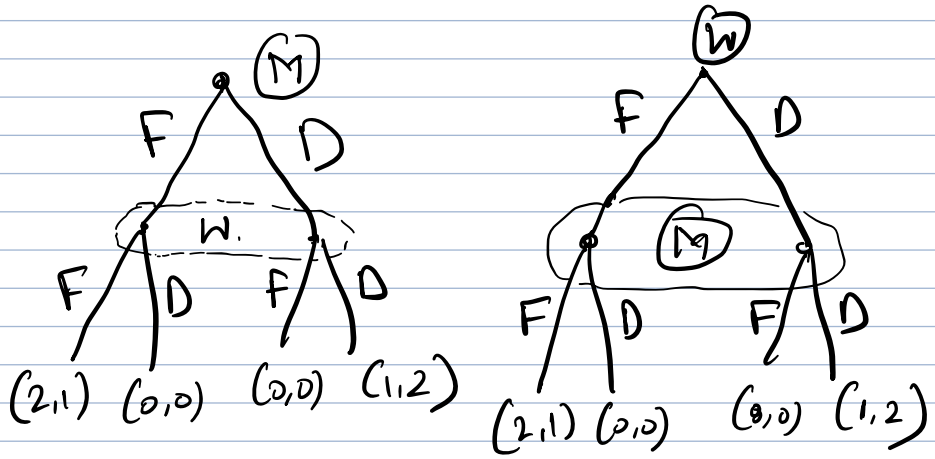
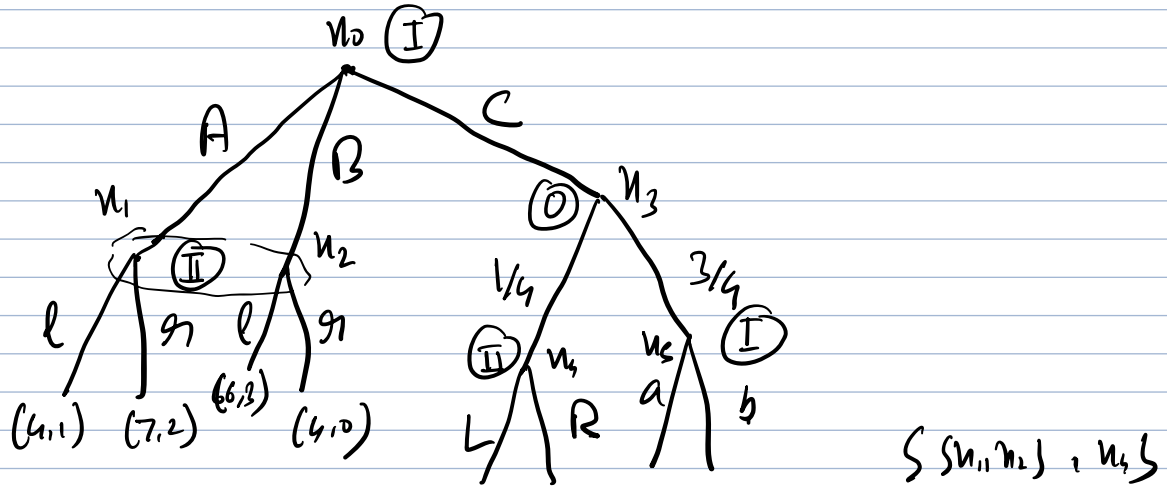
Class 22

Extensive form games with imperfect information





A game in extensive form (with imperfect information and chance moves) is a tuple

$$\Gamma = \langle N, V, E, n_0, (V_i)_{i \in N \cup \{0\}}, \mathcal{H} = (H_i)_{i \in N}, A = (A(h))_{h \in \bigcup_i H_i}, (p_n)_{n \in V_0}, (u_i)_{i \in N} \rangle$$

- N: the set of players
- (V, E, n_0): a directed rooted tree.
- $(V_i)_{i \in N \cup \{0\}}$: a partition of the set of non-leaf nodes among the agents and nature.

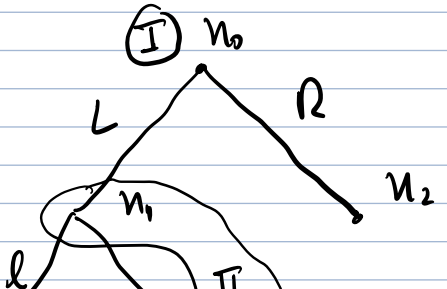
- $\mathcal{H} = (H_i)_{i \in \mathbb{N}}$: H_i is a partition of V_i. Each set in the partition is called an information set. Each $h \in H_i$ must satisfy
  (i) each node in h has the same number of outgoing edges.
  (ii) every path in (V, E, n_0) intersects h at most once.

- $\mathcal{A} = (A(h))_{h \in \bigcup_{i=1}^{n} H_i}$: for each $h \in \bigcup_{i=1}^{n} H_i$, $A(h)$ is a partition of edges coming out from the nodes in h such that for all $a \in A(h)$ contains exactly one edge outgoing from n for all $n \in h$.

  A set $a \in A(h)$ is called an action at h.

- $(p_n)_{n \in V_0}$: p_n denotes a probability distribution over the edges outgoing from n.

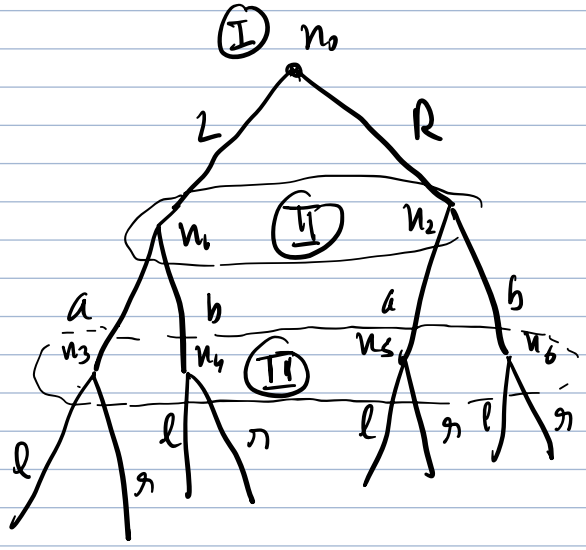
- $(u_i)_{i \in \mathbb{N}}$: the utility functions defined at the leaf nodes.





(V, E, n_0)

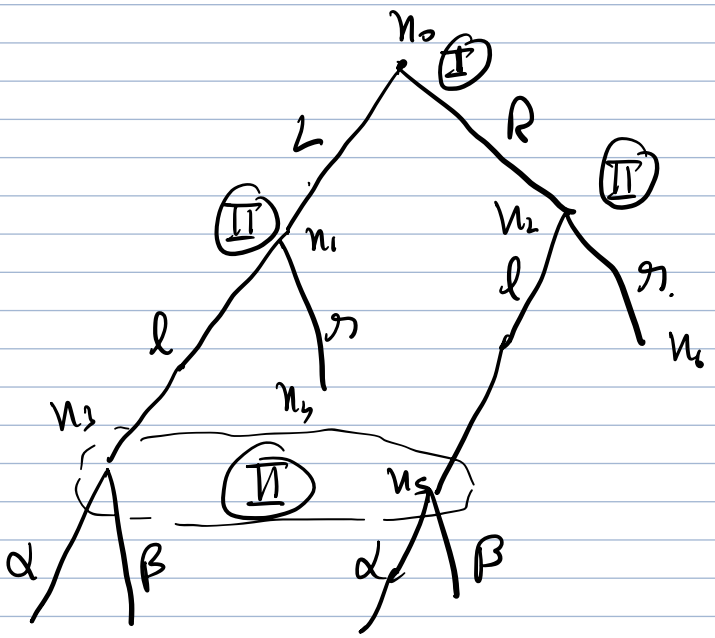
Not a valid extensive form game.



Valid extensive form game.

(does not satisfy perfect recall)

Defn (Perfect recall) An extensive form game satisfies perfect recall for player i if for all information set $h \in H_i$ and for all $u, v \in h$, an action of Player i is on the path from root to u iff it is on the path from root to v.



This game violates perfect recall.